Day 22: Binary Logistic Regression Review

Today we'll read a published study that uses a multiple logistic regression model, write out the model the authors used, and fit it on their data. We have two main questions: what is this model actually saying, and which inference tool answers the question the authors asked?

> **The paper**
>
> Wang, P.-Y., Chen, K.-L., Yang, S.-Y., & Lin, P.-H. (2019). Relationship of sleep quality, smartphone dependence, and health-related behaviors in female junior college students. *PLOS ONE*, 14(4), e0214769. https://doi.org/10.1371/journal.pone.0214769
> Open the paper. We'll focus on the Abstract, Statistical analysis section and Table 2.

Warm up

The authors measured smartphone dependence with four items. Score yourself on each from 1 (do not agree) to 5 (completely agree), and add them up:

- I feel uneasy or anxious when my phone isn't next to me
- I find myself wanting to stare at my phone screen without meaning to
- The first thing I do in the morning is check my phone for messages
- I keep checking my phone for messages during class

Your total will be between 4 and 20. Make a note of your total here. Keep it to yourself; you'll use it in Part 3.

Part 1: Read the paper

Start by reading the **Abstract** and the **Statistical Analysis** section, then skim **Table 2.**

Q1. Based on your reading, what is the response variable and what are the predictor variables? What does it mean if $Y = 1$? What about $Y = 0$?

Q2. The PSQI is a score from 0 to 21, and the authors cut it into two groups. What do they gain by doing

this? What information do they lose?

Q3. Write out the model on the log-odds scale, using π_i. Say what each x is. Which β goes with the odds ratio of 0.88 (top row in Table 2)?

Q4. Your model doesn't have an ϵ_i in it. Consider two students with exactly the same age, BMI, smoking, drinking, phone use, and dependence score. What does the model say is the same for these two students? Does it say they will have the same sleep quality?

Q5. Interpret the odds ratio of 0.88 in context. Why is it below 1? (*Careful:* which outcome is the model for?)

Q6. Given the odds ratio is 0.88, what is $\widehat{\beta}$? Which predictor goes with this value?

Part 2: Fit their model

The authors posted their data, so we can check their work! Open the R activity on your computer and follow the instructions there. Keep track of your written questions here.

Q7. Fill in the blanks to fit the model you wrote in Q3. Did you get the same $\widehat{\beta}$ as the paper?

Part 3: What is the model saying?

Each student has $Y = 0$ or $Y = 1$, so the model isn't trying to predict those points directly. It is describing the **probability** of good sleep. To see this, we'll compare the observed proportion of good sleepers at each dependence score to the fitted curve for a typical student (17 years old, BMI of 20, doesn't smoke or drink, uses a phone before bed).

Q8. Make the aggregated plot in R. Which points should we pay the least attention to, and why? Does the curve describe the overall trend?

Q9. This curve looks close to a straight line. Change `4:20` to `-40:60` in the code above and run it again (those scores don't exist, but it shows us the math). What shape is it now? Why did it look like a line over the real scores?

Q10. Next, let's look at the fitted model on all three scales for a "typical" student. Each row is 4 points higher than the row before it. Going from one row to the next:

	phone_dependence	log_odds	odds	prob
1	8	0.49566112	1.6415832	0.6214391
2	12	-0.02048906	0.9797194	0.4948779
3	16	-0.53663924	0.5847100	0.3689697
4	20	-1.05278943	0.3489630	0.2586898

a. What do you **add** to get the next log-odds? Which number in your coefficient table does this come from?

b. What do you **multiply** by to get the next odds?

c. Is there one number you can add or multiply by to get the next probability?

Q11. Use `predict()` to find the predicted probability of good sleep for a typical student with the score you gave yourself in the warm up. Should you believe this number about yourself? Who was in this study?

Q12. In their results, the authors describe "a greater the degree of smartphone dependence associated with shorter sleep length" (Wang et al., 2019). Does your model support this exact statement? How would you rewrite it?

Part 4: Which test answers the question?

We now have two tools: the z-test and CI for a coefficient, and the LRT for nested models. Notice that the paper reports an odds ratio and CI rather than an LRT.

Q13. Write the null hypothesis, in terms of the βs in your model, that matches each question:

a. After accounting for the other variables, is smartphone dependence associated with sleep quality?

b. Do any of the six predictors help us explain sleep quality?

Which of these is the question the authors are asking?

The overall LRT

For this section, you can run your own tests in R or use the output here.

```
Analysis of Deviance Table

Model 1: good_sleep ~ 1
Model 2: good_sleep ~ phone_dependence + age + bmi + smoking + drinking +
    phone_before_bed
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1       408     566.29
2       402     542.10  6   24.183 0.0004833 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Q14. What can you conclude from the test above? Based on this test alone, what can you say about smartphone dependence?

Q15. Now fit the model with everything *except* phone_dependence, and do two more LRTs:

```
Analysis of Deviance Table

Model 1: good_sleep ~ 1
Model 2: good_sleep ~ age + bmi + smoking + drinking + phone_before_bed
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       408     566.29
2       403     558.78  5   7.5083   0.1855
```

```
Analysis of Deviance Table

Model 1: good_sleep ~ age + bmi + smoking + drinking + phone_before_bed
Model 2: good_sleep ~ phone_dependence + age + bmi + smoking + drinking +
    phone_before_bed
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1       403     558.78
2       402     542.10  1   16.675 4.437e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do the two Λ values relate to Λ from the overall LRT? What was driving the significant overall test? Why would it be misleading to report only the overall test?

Q16. Pretend you're telling a friend about this result. You have the `anova()` output and the coefficient table with its CI. What can you tell them using the coefficient table that you can't tell them using the `anova()` output? Write the sentence you would say, using the odds of good sleep and a **4-point** higher dependence score.

Q17. Fill in this table:

If your question is about…	Hypothesis	Use	What you get
one predictor, after accounting for the others			
a group of coefficients (several related predictors, a categorical variable with > 2 levels, an interaction)			
whether anything in the model helps			

Q18. In 2-3 sentences: why do you think papers like this one mostly report odds ratios and CIs for individual coefficients rather than LRTs?

Your turn

Using the original score. The authors could have kept the PSQI (`psqi`) as a quantitative response. Fit a linear regression of `psqi` on the same six predictors.

- What is the sign of the `phone_dependence` slope? Why is it the opposite of the sign in `sleep_mod`?
- Interpret the slope. Do the two models lead you to the same overall conclusion about smartphone dependence?

Function quick reference

Function	Purpose
`glm(y ~ x, data, family = "binomial")`	Fit a binary logistic regression model
`tidy(model)`	Coefficient table with z-tests
`predict(model, newdata, type = "response")`	Predicted probabilities (leave out `type` for log-odds)
`anova(reduced, full, test = "LRT")`	Likelihood ratio test comparing two nested models
`cor()`	Correlation matrix for a set of quantitative variables